

Full Examination

July 18, 2025

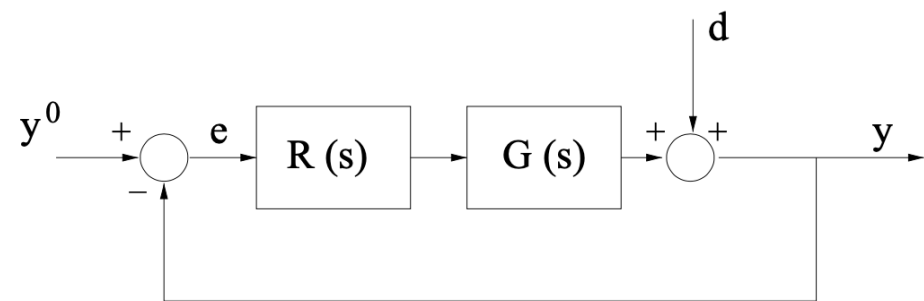
Part 2 - Design of a Controller

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Description of the Feedback Control System

Consider the feedback control system described by the block-scheme shown in the following figure



where $G(s)$ is given by

$$G(s) = \frac{1}{25} \frac{(1 + 3s) \left(1 + \frac{s}{275}\right)}{\left(1 + \frac{1}{50}s + \frac{s^2}{400}\right) \left(1 + \frac{s}{297}\right)}$$

Question

Q1: Design a **proper controller** $C(s) = \mu_C \frac{1 + sT}{s}$, $\mu_C > 0$, $T > 0$ such that all following requirements are simultaneously met:

- Zero steady-state error in unit step response (closed-loop) $e(\infty) = 0$, when $d(t) = 0 \ \forall t$

$$y^o(t) = 1(t) \implies e_\infty = 0$$

where $\mathbf{1}(t)$ denotes the unit step function;

- phase margin greater than or equal to 60° : $\varphi_m \geq 60^\circ$;
- For $d(t) = D \sin(\omega t) \cdot \mathbf{1}(t)$, $\forall D \in \mathbb{R}$, $D > 0$, $\omega \in [4, 28]$ rad/s, the corresponding steady-state output $y_{d,\infty}(t)$ generated by the disturbance $d(t)$, with $y^o(t) = 0 \ \forall t$ has magnitude satisfying $|y_{d,\infty}(t)| \leq D/120$;

Hint

How to handle the requirement on the disturbance $d(t)$? Recall the analysis we performed in Part 9 of the material (**Part 9, p. 48, 59**), in Part 10 (**Part 10, p. 5**), and the example [DESIGN_REQS_D_N](#), available in the file repository part of the MS Teams course team.